

other through satellite in a single hop. VSATs operate in Ku band, although there exist some C-band systems.

Satellite-terrestrial hybrid communication

Satellite Internet services are used in locations where terrestrial Internet access is not available and in locations which move frequently. Internet access via satellite is available worldwide, including vessels at sea and mobile land vehicles.

Two-way satellite Internet service involves both sending and receiving data from the remote VSAT site via satellite to a hub teleport, which then relays data via the terrestrial Internet. The satellite dish at each location must be precisely pointed to avoid interference with other satellites. At each VSAT site, the uplink frequency, bit rate and power must be accurately set, under control of the service provider hub.

There are several types of two-way satellite Internet services, including TDMA and SCPC. Two-way systems can be simple VSAT terminals with a 60-100cm dish and output power of only a few watts intended for consumers and small business or larger systems that provide more bandwidth. Such systems are frequently marketed as 'satellite broadband' and can cost two-three times as much per month as land-based systems such as ADSL. The modems required for this service are often proprietary, but some are compatible with several different providers. They are also expensive, costing in the range of Rs 30,000 to 100,000.

Home users tend to make use of shared satellite capacity, to reduce the cost, while still allowing high peak bit rates when congestion is absent. There are usually restrictive time-based bandwidth allowances so that users get their fair share, according to their payment.

Evolution in technology for space

Over the last four decades, all technologies related to satellite communica-

tion evolved and changed considerably resulting in more powerful and reliable satellites. In the area of communication payload, the EIRP and G/T have increased, facilitating use of small and low-cost ground terminals. Having used up the C-band spectrum, Ku-band frequency is used increasingly throughout the world.

Advances in spacecraft antenna have resulted in shaped beam antennae avoiding wastage of onboard power, which is premium in a satellite. Lightweight dual-grid antennae are used for combining transmit and receive functions on a single antenna. Spot beams are used for illuminating the desired areas with area-specific contents. Additionally, multiple spot beams facilitate frequency reuse, thus enabling effective use of the radio spectrum.

Satellite onboard batteries too have evolved considerably. The technology has changed from NiCd to NiH and now to lithium-ion batteries. The size and weight advantages have been gained. Lithium-ion batteries are also very attractive from housekeeping point of view, improving the life of the battery.

Solar cells that go into the solar array generating power onboard the satellite have advanced considerably over the years. Today, we have more efficient space-worthy solar cells. The evolution has been from Si cells of early years to GaAs cells to dual-junction GaAs to triple-junction and advanced triple-junction cells now. From the initial years to now, the sun efficiency of the solar cells has doubled, resulting in higher power generation onboard.

With all these advances and availability of powerful launchers, the capability of current communication satellites has increased enormously. Today, we can have satellites with up to 150 transponders and life of 15 years. Regenerative transponders have come into existence, enabling improved signal-to-noise ratio for communications and onboard switching of links.

The progressions of Intelsat over four decades to reach the latest Intelsat 20 that will be launched in 2012

with 74 transponders and a life of 15 years, and of INSAT over four generations, are quite impressive. So is the case with other global and regional systems such as INMARSAT and EUTELSAT. The ground system technology associated with satellite communications has advanced and the information throughput of a standard 40MHz satellite transponder has increased remarkably. In 1975, the SITE transponder of 40MHz bandwidth carried just one TV channel and today, in 2010, capability exists to broadcast eleven TV channels over the same satellite bandwidth.

Future trends

In the years to come, use of very small-aperture terminals for business and rural applications is expected to grow further in all parts of the world. Residential and rural VSATs are expected to provide broadband services with a significant reduction in tariff.

A new era for satellite mobile communications will start. Future mobile satellite systems will provide mobile video, navigation and emergency assistance services.

Use of DTH broadcasting is also expected to increase in many parts of the world, though in certain areas satellite broadcasting system will have to compete with other television delivery modes. However, direct broadcast will complement the well-established delivery systems, in some cases.

Use of the present under-utilised radio frequency bands such as 20/30 GHz will go up. Mesh networking or business-to-business connectivity without doing double hop will result in better user experience and reduction of bandwidth cost by half. In order to realise this, space-hardened routers will be integrated onboard the satellite. Satellite and ground segments will be merged to optimise network performance while enhancing service options. ●

The author is former deputy director (Satellite Communications), ISRO Headquarters, Bengaluru